

Aayush Shelar

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EDUCATION

- Sharad Institute Of Technology College Of Engineering** Maharashtra, India
Bachelor of Technology - Electronic and Computer Engineering; GPA: 6.73 July 2022 - Present
Courses: Operating Systems, Data Structures, Artificial Intelligence, Computer Network, DBMS, Embedded System

SKILLS SUMMARY

- Languages:** Python, C++, JavaScript, SQL, JAVA
- Frameworks:** Scikit-learn, TensorFlow, PyTorch, NumPy, NodeJS
- Tools:** Jupyter Notebook, Docker, GIT, Kaggle Kernels, MySQL, Google Colab
- Platforms:** Web, Windows, Arduino, Raspberry, AWS, GCP
- Soft Skills:** Leadership, Event Management, Collaboration, Responsible for time management and problem-solving

EXPERIENCE

- Rapid System** Ichalkarnji
Student Developer (Intern) Jun 2024 - July 2024
 - Worked on microcontroller-based and embedded systems:** Assisted in the development, troubleshooting, or testing of embedded software solutions and microcontroller boards used in dynamic balancing machines, gaining hands-on experience with real industrial hardware.
 - Worked with industry professionals to solve technical problems:** Participated in team work and communication with experienced engineers, improving practical knowledge in industrial electronics and embedded programming while supporting service and maintenance activities. These points show your technical contribution and industry exposure during the internship

PROJECTS

- MAITRI (Multimodal AI for Therapeutic Interactions)** : An offline AI companion for astronaut psychological support, designed for the Smart India Hackathon (ISRO). **Tech:** Python, OpenCV, TensorFlow Lite, PyTorch, Librosa, NLP (Distilled BlenderBot), Docker, ONNX Runtime. **Description:** Developed an end-to-end, fully offline system to monitor and support astronaut well-being. The system uses a lightweight CNN (MobileNetV2) for real-time facial emotion detection and a GRU model for voice-based emotion recognition. It integrates a generative dialogue agent to provide context-aware, therapeutic conversations and evidence-based interventions (CBT, mindfulness), demonstrating expertise in multimodal AI, model optimization, and edge deployment. (November '25)
- AI-Powered Predictive Traffic Flow & Anomaly Detector:** A smart urban mobility solution for proactive traffic management through predictive analytics. **Tech:** Python, OpenCV, Ultralytics YOLOv8, Pandas, Scikit-learn (Isolation Forest), NumPy, Matplotlib. **Description:** Built a full-cycle ML system that analyzes a video stream to monitor traffic flow. The project extracts time-series data (vehicle count) in real-time, uses a simple forecasting model to predict future congestion, and employs an Isolation Forest model to detect traffic anomalies. This showcases proficiency in turning unstructured video data into actionable insights for urban planning. (August '24)
- Financial Market News Sentiment Analysis:** A sentiment analysis pipeline to translate financial news into actionable insights for investors and market analysts. **Tech:** Natural Language Processing (NLP) techniques, Python (or specific libraries like NLTK, spaCy, or Hugging Face). **Description:** Designed and implemented a sentiment analysis system to process financial news articles. The pipeline utilizes NLP techniques to extract sentiment scores, providing a quantitative basis for assessing market sentiment and supporting data-driven investment and trend analysis decisions. (March '23)
- IoT-Based Underground Cable Fault Detection System:** A real-time, automated system for detecting and locating faults in underground power cables to minimize downtime and enhance grid reliability. **Tech:** Atmel (ATmega) Microcontroller, ESP8266 Wi-Fi Module, GSM Module, LCD, Potential Divider Network, Cloud-based Analytics. **Description:** Developed and deployed a real-time fault detection system for underground cables. The system uses a potential divider network to precisely locate faults and displays the data on an LCD. It provides instant wireless alerts via GSM to technicians, and integrates with a cloud-based platform for continuous monitoring and predictive maintenance, thereby supporting smart city infrastructure goals. (September '25)

HONORS AND AWARDS

- Participated at DIPEX - a State Level Exhibition cum Competition of Working Models - March, 2025
- Runner's Up at Aquaquest of Electrovert a International Level Mega Event - November, 2024

VOLUNTEER EXPERIENCE

- Event Head at Developer Student Clubs ECESA** Maharashtra, India
Conducted online and offline training & a Code Trix Event impacting over 1500 students. Jan 2024 - Present
- Volunteer at ETESA with Team TRINETRA in IEETE** Maharashtra, India
Organized events, conducted workshops and delivered workshops reaching over 250 students. Jan 2023